

## Lab 1: Experiencing Blocking (Row-Level Lock)

Query

Query History

1

-- T1

2

-- request X lock

3

BEGIN;

4

SELECT film\_id,rental\_rate

5

FROM film WHERE film\_id = 1 FOR UPDATE;

6

7

8

9

UPDATE film SET rental\_rate = 1

10

WHERE film\_id =1;

11

COMMIT;

Data Output

Messages

Notifications

Showing rows: 1 to 1

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| film_id      | rental_rate   |
|--------------|---------------|
| [PK] integer | numeric (4,2) |
| 1            | 2.00          |

Total rows: 1

Query complete 00:00:00.157

CRLF

Ln 4, Col 1

Query

Query History

1

-- T2

2

3

4

BEGIN;

5

-- request a X lock (FOR UPDATE)

6

SELECT film\_id,rental\_rate

7

FROM film WHERE film\_id = 1 FOR UPDATE;

8

9

10

--

11

UPDATE film SET rental\_rate = 2

12

WHERE film\_id = 1

13

COMMIT;

Data Output

Messages

Notifications

Waiting for the query to complete... 00:00:16.325

CRLF

Ln 6, Col 1

Query

Query History

1

-- T1

2

-- request X lock

3

BEGIN;

4

SELECT film\_id,rental\_rate

5

FROM film WHERE film\_id = 1 FOR UPDATE;

6

7

8

9

UPDATE film SET rental\_rate = 1

10

WHERE film\_id =1;

11

COMMIT;

Data Output

Messages

Notifications

COMMIT

Query returned successfully in 59 msec.

Total rows: 1

Query complete 00:00:00.059

CRLF

Ln 11, Col 1

Query

Query History

1

-- T2

2

3

4

BEGIN;

5

-- request a X lock (FOR UPDATE)

6

SELECT film\_id,rental\_rate

7

FROM film WHERE film\_id = 1 FOR UPDATE;

8

9

10

--

11

UPDATE film SET rental\_rate = 2

12

WHERE film\_id = 1

13

COMMIT;

Data Output

Messages

Notifications

Waiting for the query to complete... 00:01:50.873

CRLF

Ln 6, Col 1

Query

Query History

1

-- T1

2

-- request X lock

3

BEGIN;

4

SELECT film\_id,rental\_rate

5

FROM film WHERE film\_id = 1 FOR UPDATE;

6

7

8

9

UPDATE film SET rental\_rate = 1

10

WHERE film\_id =1;

11

COMMIT;

Data Output

Messages

Notifications

UPDATE 1

Query returned successfully in 5 secs 161 msec.

Total rows: 1

Query complete 00:00:05.161

CRLF

Ln 3, Col 1

Query

Query History

1

-- T2

2

3

4

BEGIN;

5

-- request a X lock (FOR UPDATE)

6

SELECT film\_id,rental\_rate

7

FROM film WHERE film\_id = 1 FOR UPDATE;

8

9

10

--

11

UPDATE film SET rental\_rate = 2

12

WHERE film\_id = 1

13

COMMIT;

Data Output

Messages

Notifications

Waiting for the query to complete... 00:00:03.235

CRLF

Ln 4, Col 1

| Query                                                                                                                                                                                                    | Query History | Query                                                                                                                                                                                                                      | Query History |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <pre> 1  -- T1 2  -- request X lock 3  BEGIN; 4  SELECT film_id,rental_rate 5  FROM film WHERE film_id = 1 FOR UPDATE; 6 7 8 9 10 UPDATE film SET rental_rate = 1 11 WHERE film_id =1; 12 COMMIT; </pre> |               | <pre> 1  -- T2 2 3 4  BEGIN; 5  -- request a X lock (FOR UPDATE) 6  SELECT film_id,rental_rate 7  FROM film WHERE film_id = 1 FOR UPDATE; 8 9 10 11 UPDATE film SET rental_rate = 2 12 WHERE film_id = 1 13 COMMIT; </pre> |               |
| <p>Data Output Messages Notifications</p> <p>COMMIT</p> <p>Query returned successfully in 47 msec.</p>                                                                                                   |               | <p>Data Output Messages Notifications</p> <p>UPDATE 1</p> <p>Query returned successfully in 30 secs 468 msec.</p>                                                                                                          |               |
| <p>Total rows: Querv complete 00:00:00.047 CRLF Ln 11. Col 1</p>                                                                                                                                         |               | <p>Total rows: Querv complete 00:00:30.468 CRLF Ln 4. Col 1</p>                                                                                                                                                            |               |

## Lab 2: Replicating Non-repeatable Read

| Query                                                                                                                                                                                                                                   | Query History | Query                                                                                                           | Query History |  |  |                                                                                                          |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|---------------|--|--|----------------------------------------------------------------------------------------------------------|--|
| <pre> 1  -- Step 1 2  BEGIN; 3  select rental_rate from film 4  where film_id = 10; 5  -- Step 3 6  select rental_rate from film 7  where film_id = 10; 8  COMMIT; </pre>                                                               |               | <pre> 1  -- Step 2 2 3 4  BEGIN; 5  UPDATE film SET rental_rate = 0.99 6  WHERE film_id = 10; 7  COMMIT; </pre> |               |  |  |                                                                                                          |  |
| <p>Data Output Messages Notifications</p> <p>Showing rows: 1 to 1</p> <p>Page No: 1 of 1</p> <table border="1"> <thead> <tr> <th>rental_rate</th> <th></th> </tr> </thead> <tbody> <tr> <td>4.99</td> <td></td> </tr> </tbody> </table> | rental_rate   |                                                                                                                 | 4.99          |  |  | <p>Data Output Messages Notifications</p> <p>ROLLBACK</p> <p>Query returned successfully in 57 msec.</p> |  |
| rental_rate                                                                                                                                                                                                                             |               |                                                                                                                 |               |  |  |                                                                                                          |  |
| 4.99                                                                                                                                                                                                                                    |               |                                                                                                                 |               |  |  |                                                                                                          |  |

Query

Query History

```

1  -- Step 1
2  BEGIN;
3  select rental_rate from film
4  where film_id = 10;
5  -- Step 3
6  select rental_rate from film
7  where film_id = 10;
8  COMMIT;

```

Query

Query History

```

1  -- Step 2
2
3
4  BEGIN;
5  UPDATE film SET rental_rate = 0.99
6  WHERE film_id = 10;
7
8  COMMIT;

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1

Page No: 1 of 1

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| rental_rate | numeric (4,2) |
|-------------|---------------|
| 1           | 0.99          |

Total rows: 1    Query complete 00:00:00.170    CRLF    Ln 6, Col 1

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1

Page No: 1 of 1

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WARNING: there is already a transaction in progress  
COMMIT

Query returned successfully in 66 msec.

Total rows:    Query complete 00:00:00.066    CRLF    Ln 4, Col 1

### Lab 3: Creating a Deadlock

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dvdrental/postgres@postgres

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Query

Query History

```

1  -- Step 1
2  BEGIN;
3  UPDATE film SET rental_rate = 2.99
4  WHERE film_id = 100;
5  -- Step 3
6  UPDATE film SET rental_rate = 3.99
7  WHERE film_id = 200;
8  COMMIT;

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1

Page No: 1 of 1

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UPDATE 1

Query returned successfully in 6 secs 994 msec.

Total rows:    Query complete 00:00:06.994

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dvdrental/postgres@postgres

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No limit

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Query

Query History

```

1  -- Step 2
2
3
4  BEGIN;
5  UPDATE film SET rental_rate = 2.99
6  WHERE film_id = 200;
7
8  -- Step 4
9  UPDATE film SET rental_rate = 3.99
10 WHERE film_id = 100;
11 COMMIT;

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1

Page No: 1 of 1

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ERROR: deadlock detected  
Process 28428 waits for ShareLock on transaction 1006; blocked by process 23140.  
Process 23140 waits for ShareLock on transaction 1005; blocked by process 28428.

SQL state: 40P01  
Detail: Process 28428 waits for ShareLock on transaction 1006; blocked by process 23140.  
Process 23140 waits for ShareLock on transaction 1005; blocked by process 28428.  
Hint: See server log for query details.

Total rows:    Query complete 00:00:01.041

# Lab 4: Preventing Race Conditions (FOR UPDATE)

Query Query History

1

-- Step 1

2

BEGIN;

3

select \* from inventory

4

where inventory\_id = 10 for update;

5

-- Step 3

6

insert into rental

7

(rental\_date, inventory\_id, customer\_id, staff\_id)

8

values (NOW(), 10 , 1 , 1);

9

COMMIT;

Data Output Messages Notifications

Showing rows: 1 to 1

Page No: 1 of 1

inventory\_id [PK] integer

film\_id smallint

store\_id smallint

last\_update timestamp without time zone

1

10

2

2

2006-02-15 10:09:17

Total rows: 1 Query complete 00:00:00.144 CRLF Ln 2, Col 1

Query Query History

1

-- Step 2

2

3

4

BEGIN;

5

select \* from inventory

6

where inventory\_id = 10 for update;

7

8

rollback;

Data Output Messages Notifications

WARNING: there is already a transaction in progress

COMMIT

Query returned successfully in 66 msec.

Successfully run. Total query runtime: 144 msec. 1 rows affected.

Total rows: Query complete 00:00:00.066 CRLF Ln 8, Col 10

Query Query History

1

-- Step 1

2

BEGIN;

3

select \* from inventory

4

where inventory\_id = 10 for update;

5

-- Step 3

6

insert into rental

7

(rental\_date, inventory\_id, customer\_id, staff\_id)

8

values (NOW(), 10 , 1 , 1);

9

COMMIT;

Data Output Messages Notifications

COMMIT

Query returned successfully in 82 msec.

Total rows: Query complete 00:00:00.082 CRLF Ln 9, Col 1

Query Query History

1

-- Step 2

2

3

4

BEGIN;

5

select \* from inventory

6

where inventory\_id = 10 for update;

7

-- Step 4

8

rollback;

Data Output Messages Notifications

Showing rows: 1 to 1

Page No: 1 of 1

inventory\_id [PK] integer

film\_id smallint

store\_id smallint

last\_update timestamp without time zone

1

10

2

2

2006-02-15 10:09:17

Total rows: 1 Query complete 00:00:31.540 CRLF Ln 4, Col 1